★ GOLD LAYER

Business-Ready Data Models for Analytics

SQL DATA WAREHOUSE PROJECT

SQL Server | Medallion Architecture

PURPOSE Document the layer workflow, key SQL patterns, validation checks, and implementation evidence.

1. Layer objective

The Gold layer organizes validated Silver data into business-friendly models for analytics, dashboards, and reporting. It should make business measures and relationships clear to report developers and decision-makers.

★ 2. Dimensional model overview

Model object	Purpose	Examples to verify
Fact table	Stores measurable business events and dimension keys	Sales, order-line, transaction
Dimension table	Stores descriptive business context	Customer, product, date, location

These are dimensional-model examples, not confirmation that these exact objects exist in the current project. Use the actual Gold DDL to complete the inventory.

3. Gold workflow

Stage	Activity	Deliverable
01	Identify business process and fact-table grain	Model definition
02	Define dimensions, keys, and attributes	Dimension design
03	Define measures and relationships	Fact design
04	Build Gold views or tables from Silver	Business-ready model
05	Validate joins, keys, and totals	Quality evidence
06	Connect reporting tools	Analytics-ready outputs

4. Modeling decisions to document

- Fact grain: explain exactly what one row represents.
- Measures: define units, revenue, cost, profit, quantity, or other implemented metrics.
- Keys: document primary keys, foreign keys, and surrogate-key strategy.
- History: specify whether dimensions use Type 1, Type 2, or another approach.
- Dates: explain role-playing dates such as order, ship, or due date when implemented.

5. Example validation queries

Replace example object and column names with the actual Gold model before execution.

```
-- Check fact rows without a matching customer dimension
SELECT COUNT(*) AS unmatched_rows
FROM gold.fact_sales AS f
LEFT JOIN gold.dim_customer AS c
    ON f.customer_key = c.customer_key
WHERE c.customer_key IS NULL;
```

```
-- Example measure total
SELECT SUM(sales_amount) AS total_sales
FROM gold.fact_sales;
```

6. Gold object inventory

Object name	Type	Grain / purpose	Silver source

[Actual Gold object]	[View/table]	[One row represents…]	[Source object]
[Actual Gold object]	[View/table]	[One row represents…]	[Source object]
[Actual Gold object]	[View/table]	[One row represents…]	[Source object]

7. Reporting and business use

- List dashboards or reports that consume Gold objects.
- Define each KPI, calculation, unit, and business meaning.
- Document available filters, dimensions, and drill-down paths.
- Attach report screenshots or links after confirming they use the final Gold objects.

✅Completion target: the model, relationships, measures, and validation results are documented and the Gold layer is ready for consistent analytical use.

 Verification note: Gold SQL scripts and model definitions were not supplied. Populate the inventory and examples with actual object names, joins, measures, and results before presenting this as completed implementation.